

Emmanuel Rodriguez

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EDUCATION

St. Mary's University | San Antonio, TX

May 2025

Bachelor of Science, Computer Engineering, GPA 3.51

SKILLS

Technical /Computer Skills: C/C++, Python, Assembly (RISC-V), GitHub, HTML, CSS, JavaScript, React, VHDL

Other: Leadership, Communication/Teamwork, Fluent in Spanish, Adaptability, Strong Work Ethic, Critical Thinking.

EXPERIENCE

RDZ Quality Homes & Construction – Laborer | San Antonio, TX

May 2023 - Present

- Assisted construction teams with plumbing, electrical, framing, flooring, and concrete through the use of several power tools and equipment to keep projects running efficiently.

SAMSAT – R&D Intern | San Antonio, TX

January 2025 – May 2025

- Restored functionality to Cyber City exhibit, a **Raspberry Pi** and OpenPLC-powered Lego city simulating a smart grid, improving visitor engagement.
- Enhanced system architecture by connecting and soldering additional **Raspberry Pi** units, increasing scalability and reliability for future interactive features.

Northern Tool & Equipment – Cashier | San Antonio, TX

August 2023 – June 2024

- Delivered exceptional customer service by managing front desk operations, handling inquiries, and resolving issues promptly.
- Accurately processed transactions and managed daily cash reconciliation, ensuring financial integrity and compliance.

TECHNICAL PROJECTS

Drone Car Automation

- Developed a **C-based control system** for manual and autonomous navigation, integrating **Raspberry Pi** remote access via VNC Viewer.
- Implemented a **curve correction algorithm** to optimize trajectory accuracy, securing 2nd place in class track race.

Camera Tracking System

- Designed a **C++ vision-based tracking application** to detect and count objects in real time, calculating velocity and angle for motion analysis.
- Built scalable architecture for storing and processing object data, showcasing advanced problem-solving and coding skills.

Web-app Portfolio

- Built and deployed several websites and web-apps, applying modern **HTML**, **CSS**, **JavaScript**, and **React** principles.
- Link: <https://e-rodriguez-portfolio.netlify.app/>

Derma-light Skin Analysis System

- Built a device to capture high-resolution skin images for early detection of potential skin cancer, using **MicroPython** to leverage the **raspberrypi pico** to speak to the app via a **TCP server** and control the camera using **SPI** connections.
- Developed a **mobile app** using **Flutter** to analyze images and securely transmit data to clinics, enhancing telehealth capabilities.

LEADERSHIP

Mariachi Cascabel | San Antonio, Tx

September 2022 – May 2025

Treasurer and Risk-Management Officer and Performer

- Supervise and delegate club finances, including managing the club funds and creating plans for budgeting.

IEEE Student Organization | San Antonio, Tx

January 2024 – May 2025

Member

- Placed 4th at IEEE Region 5 Greentech Cyber Challenge focused on Cryptography, Reverse Engineering, and more.